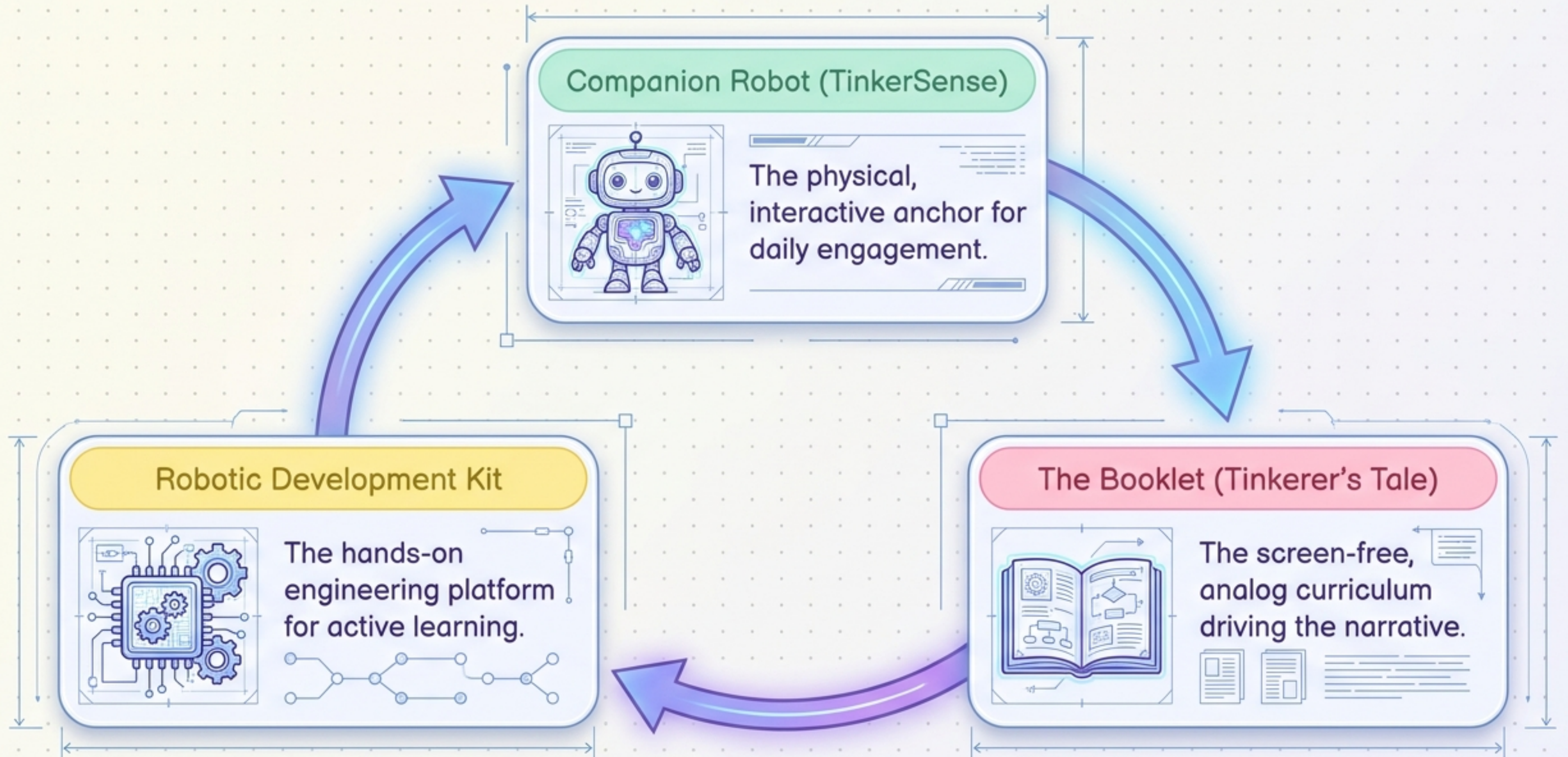


Pioneering unique human-robot interactions
specifically designed for children.

[illegible]

The TinkerSense Ecosystem



Why TinkerSense? Fostering Holistic Development



Interactive Learning

Engaging games, quizzes, and moral storytelling based on interrelated animal fables like the Panchatantra.



Behavior Nudges

Proactive reminders promoting healthy daily habits, such as handwashing and kindness.



Emotion Recognition

Advanced analysis to detect feelings and respond with appropriate empathy.



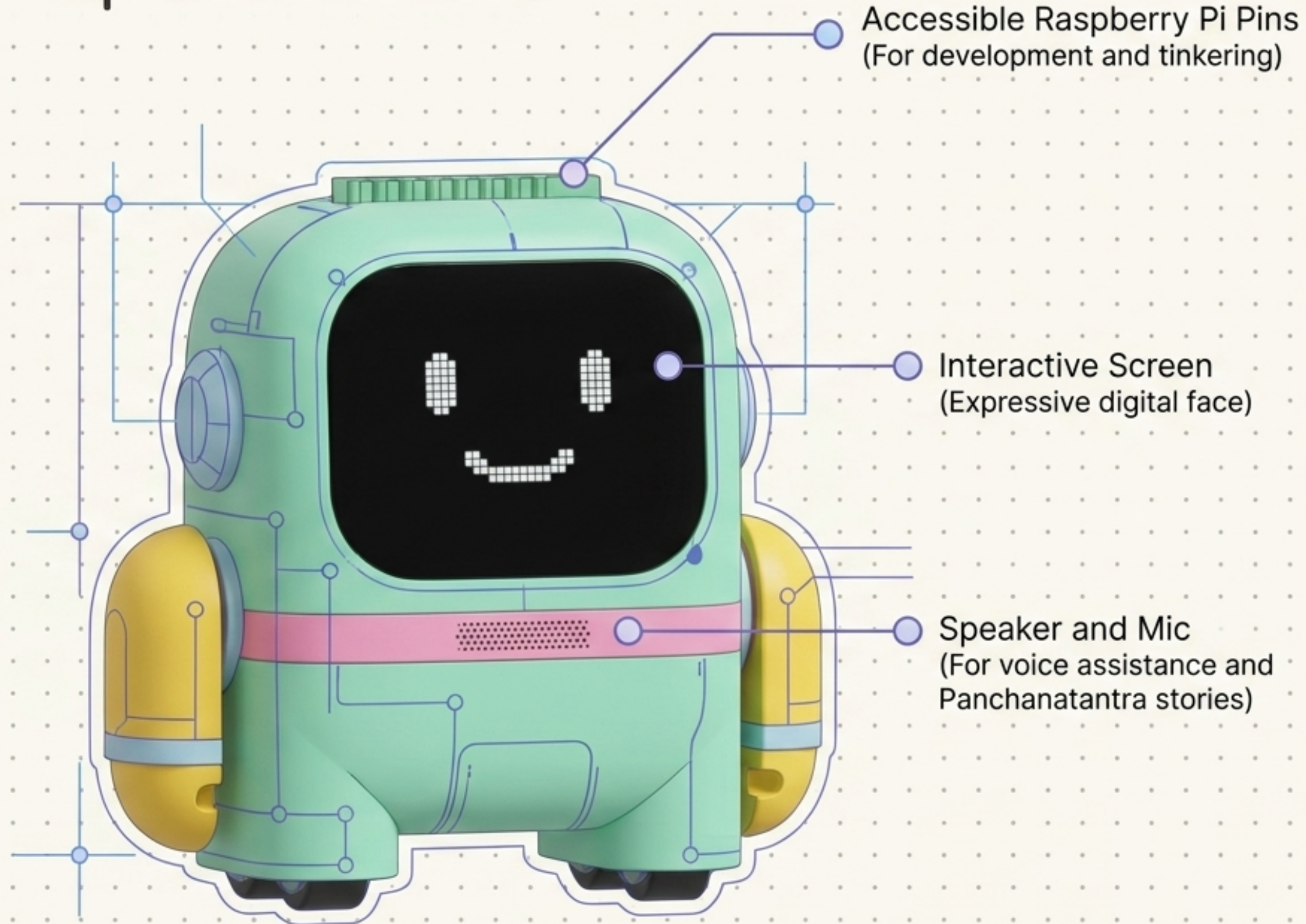
Eco-Awareness

Foundational lessons centered on climate care and sustainability.

Anatomy of the Companion Robot

Core Capabilities

- Voice assistance
- Reminders for behavior change
- Emotional & digital wellbeing tracking
- Environmental awareness



The Sensor Environment Blueprint

1. Weather Buddy: DHT11 (Temp & Humidity)
- Gives room weather & dressing advice.

2. Hand-Wash Reminder: PIR Motion Sensor - Detects entry & prompts washing.

3. Plant Watering Assistant: Soil Moisture Sensor - Alerts if a plant is dry.

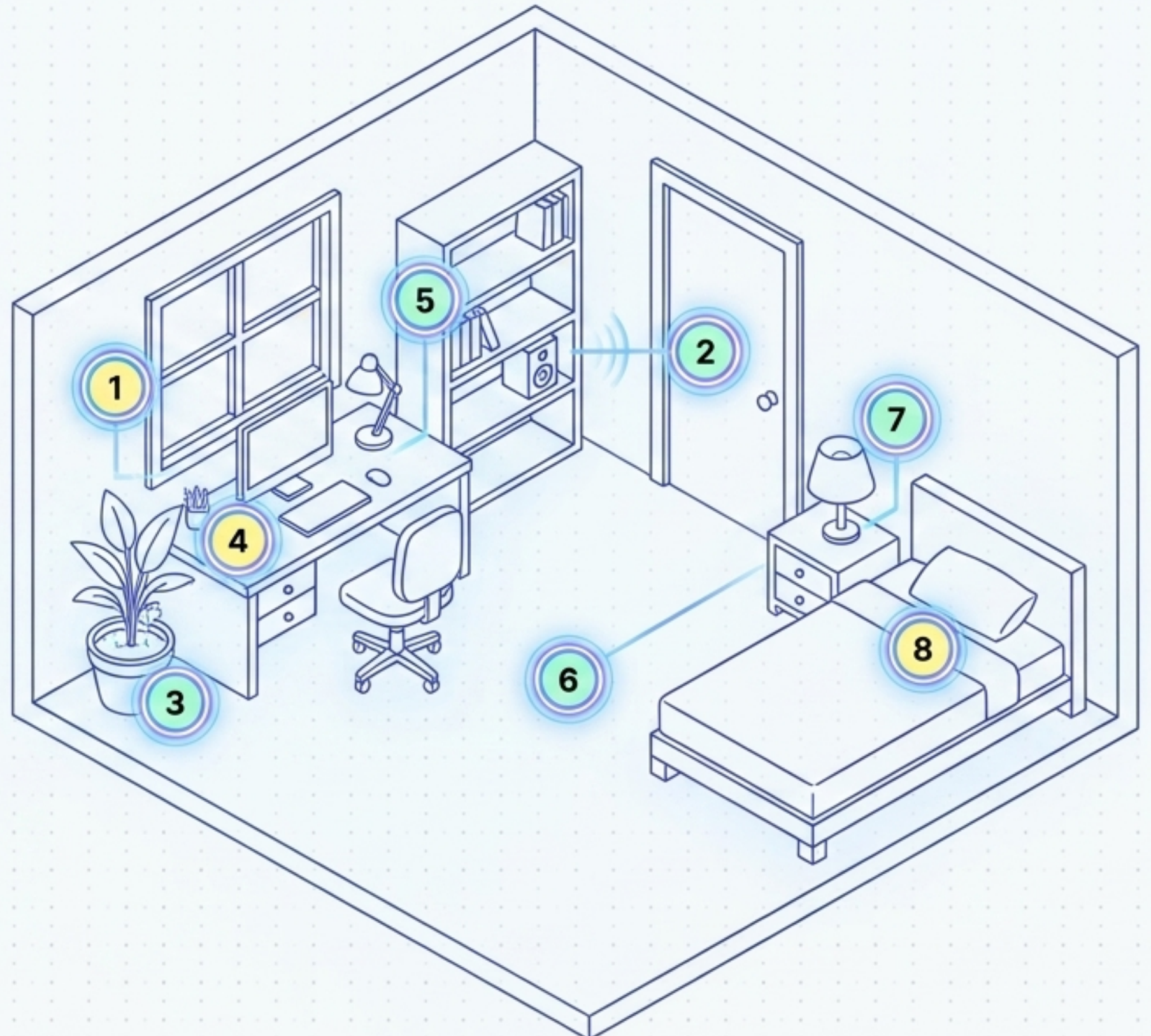
4. Light Detective: LDR - Programmed to follow the light.

5. Sound Detective: Sound Sensor - Programmed to follow sound.

6. Obstacle-Aware Robot: Ultrasonic Sensor - Avoids objects while moving.

7. Mood Light: RGB LED - Changes color corresponding to mood.

8. Clap to Talk: Sound Sensor - Activates the robot via clapping.



The Physical-Digital Bridge: Tinkerer's Tale



Adventures in the Digital World

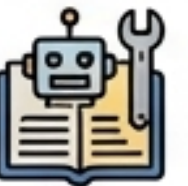
Screen-Free Complement: Balances interactive robot time with quiet reflection.



Child-Led Exploration: Includes Think & Tinker prompts to deepen independent learning and emotional expression.



Interactive & Holistic: 7 illustrated chapters featuring fun projects, reflection activities, and robot games.



Framework Aligned: Grounded in global SDGs and rigorous social-emotional learning frameworks.



A Seven-Step Developmental Journey

Chapter 1:
Robot history
& cobots

Chapter 3:
Talk to Tinker!
+ conversation games

Chapter 5:
Emotional support
& coping tools

Chapter 7:
Mindful tech use
& digital well-being

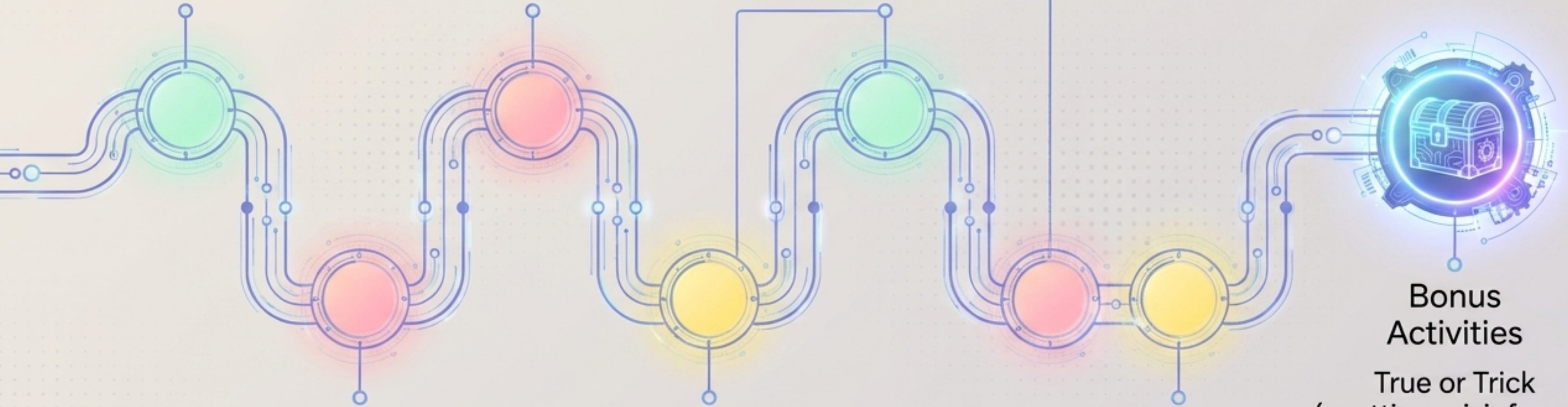
Chapter 2:
AI – deterministic
vs generative

Chapter 4:
Art, jokes, poems,
inventions

Chapter 6:
Robots saving the
planet +
eco-missions

Bonus
Activities

True or Trick
(spotting misinforma-
tion) & Tinker's
Eco Challenge
checklist.



The Baseline Brain: Hybrid AI Architecture

Dialogflow CX (Deterministic AI)

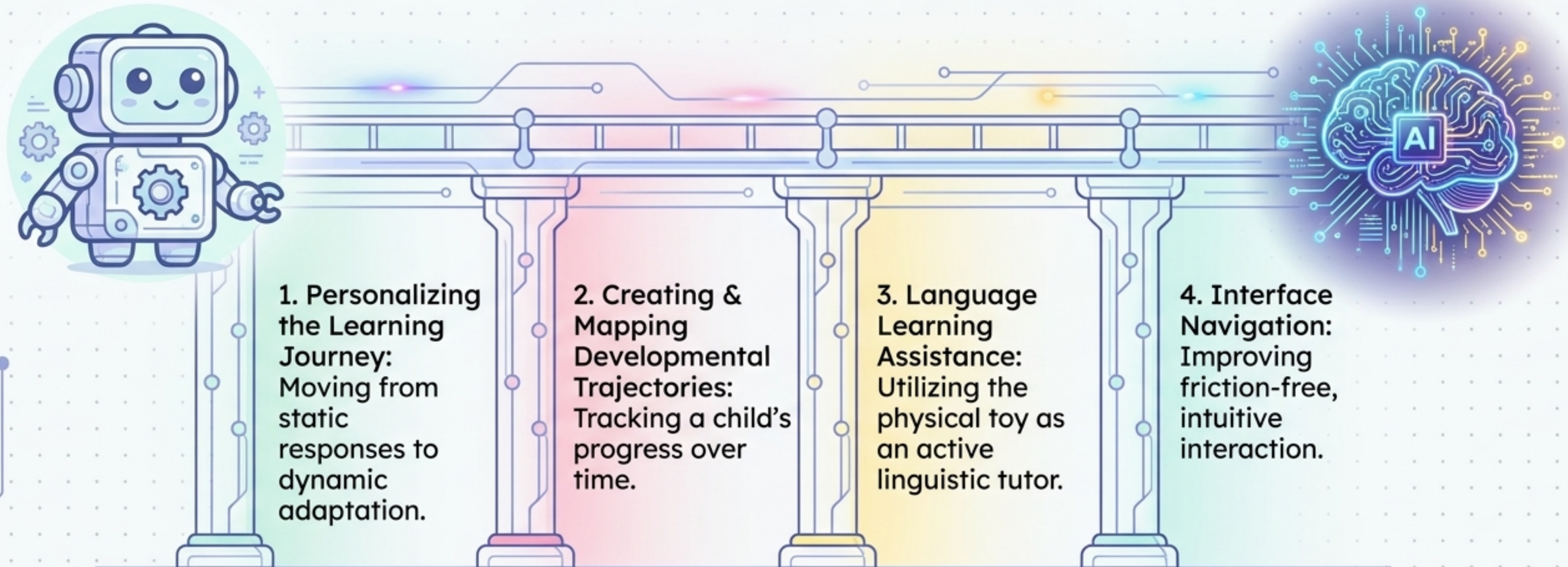
- Designed by experts for safety, structure, and factuality.
- Based directly on the general curiosity patterns of Indian children.
- Strictly age-appropriate and controlled.

Gemini 1.5 Pro (Generative AI)

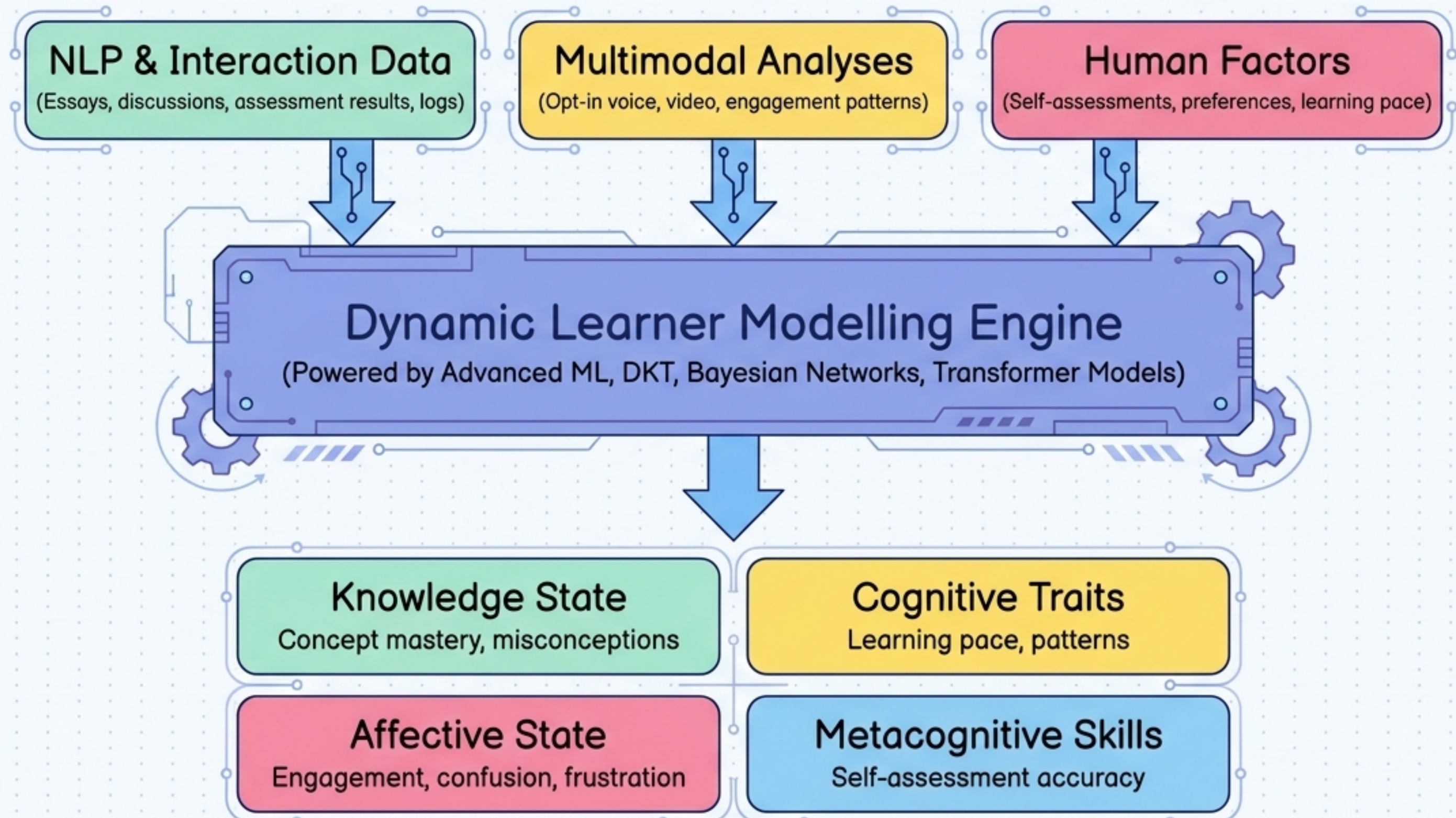
- Engineered for creativity and spontaneity.
- Handles complex open-ended questions.
- Powers dynamic, adaptive storytelling.

The Evolutionary Gap: Scope for Improvement

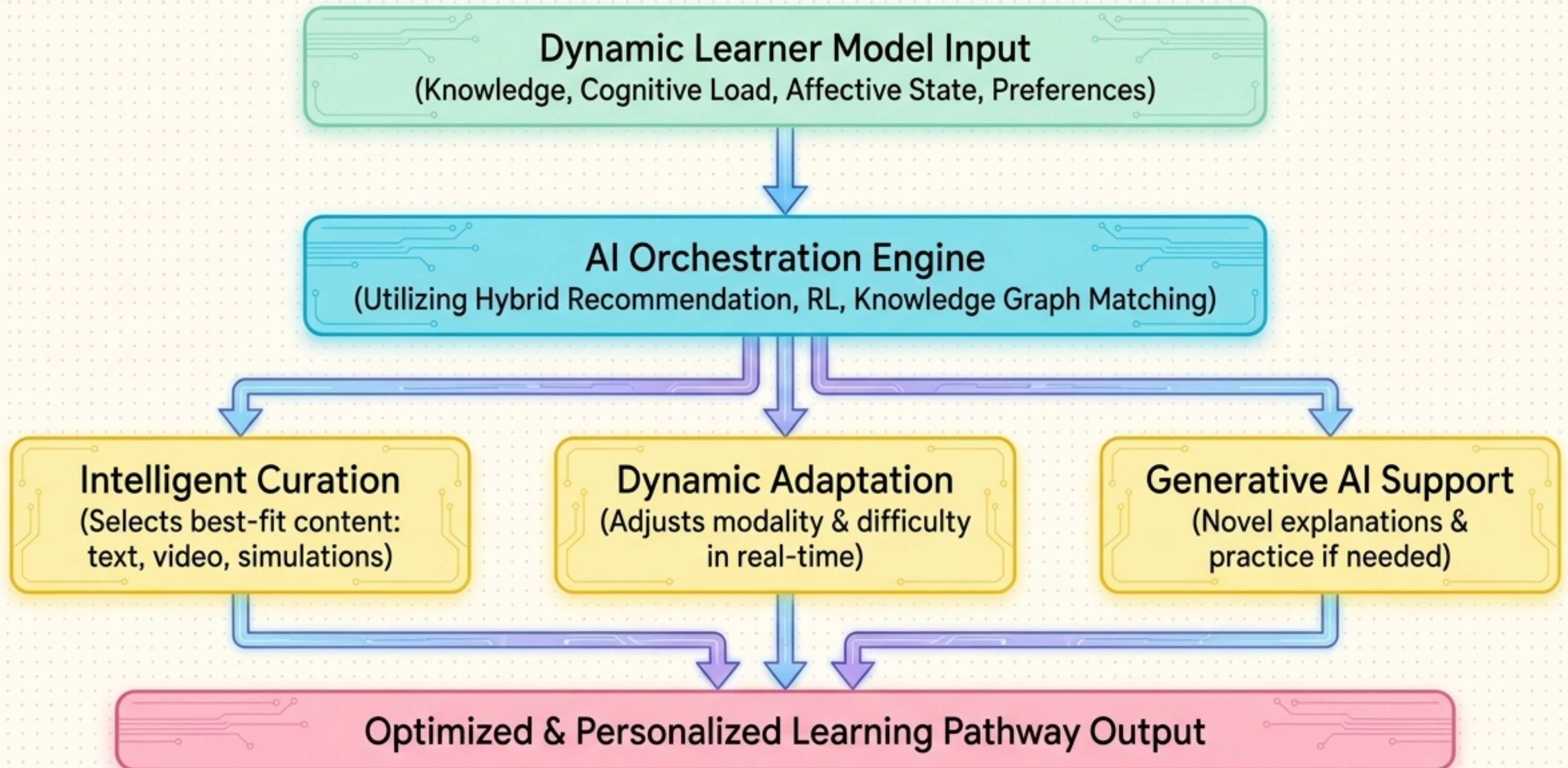
To evolve TinkerSense from a responsive toy to a closed-loop pedagogical system, four key capabilities must be built:



From Interaction to Insight: The Modelling Engine



Real-Time AI Orchestration



⚙️ The Advanced Intelligent Feedback System ⚙️

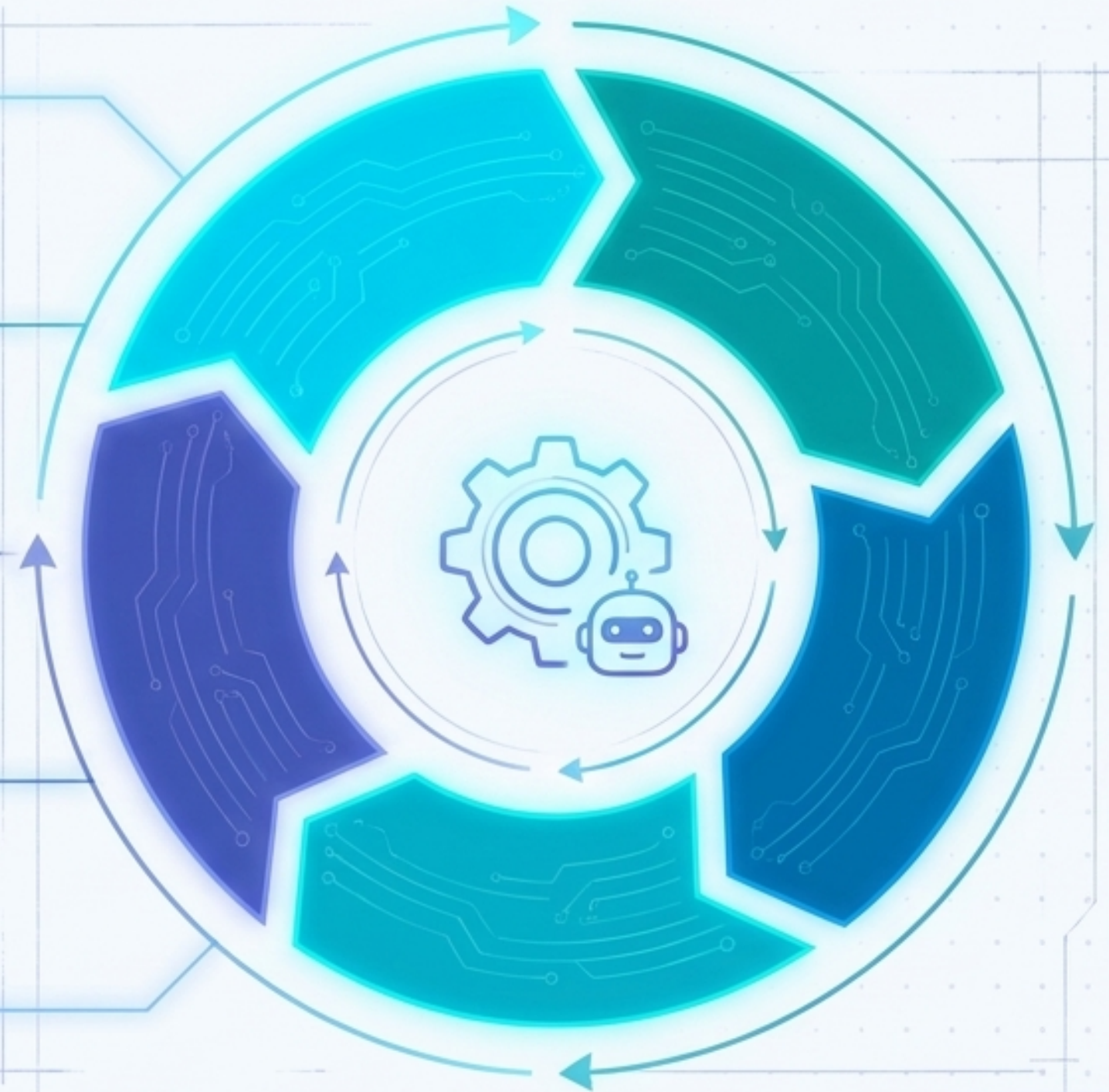
Principled LLM Integration: Carefully designed and evaluated outputs grounded by Retrieval Augmented Generation (RAG).

Theoretically Grounded Content: Feedback strictly guided by learning science (e.g., KLI framework) for memory, problem-solving, and deep understanding.

Personalized & Contextual: Tailored instantly to the student's error, learner model, and current emotional state.

Multimodal Delivery: GenAI-powered feedback delivered dynamically via text, images, audio, or short video.

Scaffolding & Metacognition: Interactive support that intentionally fades with proficiency to promote self-correction.



Development Milestones & The Road Ahead



Offline Speech-to-Text and Text-to-Speech modules successfully integrated.



Responses of the deterministic AI highly refined.



Working on line drawing of the robot for the official design patent.



Completed the first comprehensive draft of the Tinkerer's Tale booklet.

Nothing in life is to be feared, it is only to be understood. Now is the time to understand more, so that we may fear less.

– Marie Curie